Python

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Basic_Class_Prgs_Assignments

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SAMPLE PROGRAMS

==========

```
name="Hari"
age=11
print("Hello ",name,"nice to meet you and you are",age,"years old")
```

```
name=input("What is your name?")
age=input("What is your age?")
print("Hello ",name,"nice to meet you and you are",age,"years old")
```

 # WAS accept 2 numbers and find sum

```
a=input("Enter number 1 : ") # 10
b=input("Enter number 2 : ") # 20
c=a+b
print("sum=",c)  # 1020
print("a=",type(a)) # <class 'str'>
print("b=",type(b))  # <class 'str'>
```

#(or)_____

```
a=input("Enter number 1 : ") # 10
b=input("Enter number 2 : ") # 20
c=int(a)+int(b)
print("sum=",c)  # 30
print("a=",type(a)) # <class 'str'>
print("b=",type(b))  # <class 'str'>
```

#(or)_____

```
a=int(input(“Enter number 1 : “)) # 10
b=int(input(“Enter number 2 : “)) # 20
c=a+b
print(“sum=”,c)  # 30
print(“a=”,type(a)) # <class ‘int’>
print(“b=”,type(b))  # <class ‘int’>
```

```
_____________________________________________________________________
# WAP accept empno,ename,sal and calcuclate bonus based
# on the following condition
#(20% on annual_salary)
empno=int(input("Enter employee number : "))
ename=input("Enter employee name : ")
esal=float(input("Enter employee salary : "))
ann_sal=esal*12
bonus=ann_sal*20/100
print("\t\t Employee  Bonus  Report ")
print("-"  *  50)
print("\t Employee Number  :  ",empno)
print("\t Employee Name    :  ",ename)
print("\t Employee Salary  :  ",esal)
print("\n\t Annual Salary  :  ",ann_sal)
print("\t Employee Bonus   :  ",bonus)
print("-"  *  50)
_____________________________________________________________________
print("Hello")
print("Python")
print("world")
    '''
o/p is :
Hello
Python
world
'''
print("Hello",end=" ")
print("Python",end=" ")
print("world")
'''
o/p is :
Hello Python world
'''
#_____________________________________________________________________
empno=101
ename="Sai"
ejob="Manager"
```

```
esal=90000.00
deptno=10
print("Sai employee details,....",end="---> ")
print(empno,ename,ejob,esal,deptno,sep=",")
'''
o/p is :
Sai employee details,....---> 101,Sai,Manager,90000.0,10
'''
```

#_____

#Assignments

1)WAP accept product no,pname,price,qty and calculate total,disc(10% on total),and net amount and prepare bill

2) WAP accept studentname and 3 subject marks and calculate total and avg marks and prepare report

3) WAP accept empno,ename,sal and calculate ta,da,hra,total_allowences and gross salary and prepare employee pay slip

ta => 10% 0n sal

da => 15% on sal

hra => 20% on sal

total_allow => ta+da+hra

gross_sal => sal+total_allow

4) WAP accept 2 integer numbers and swap it

Operators

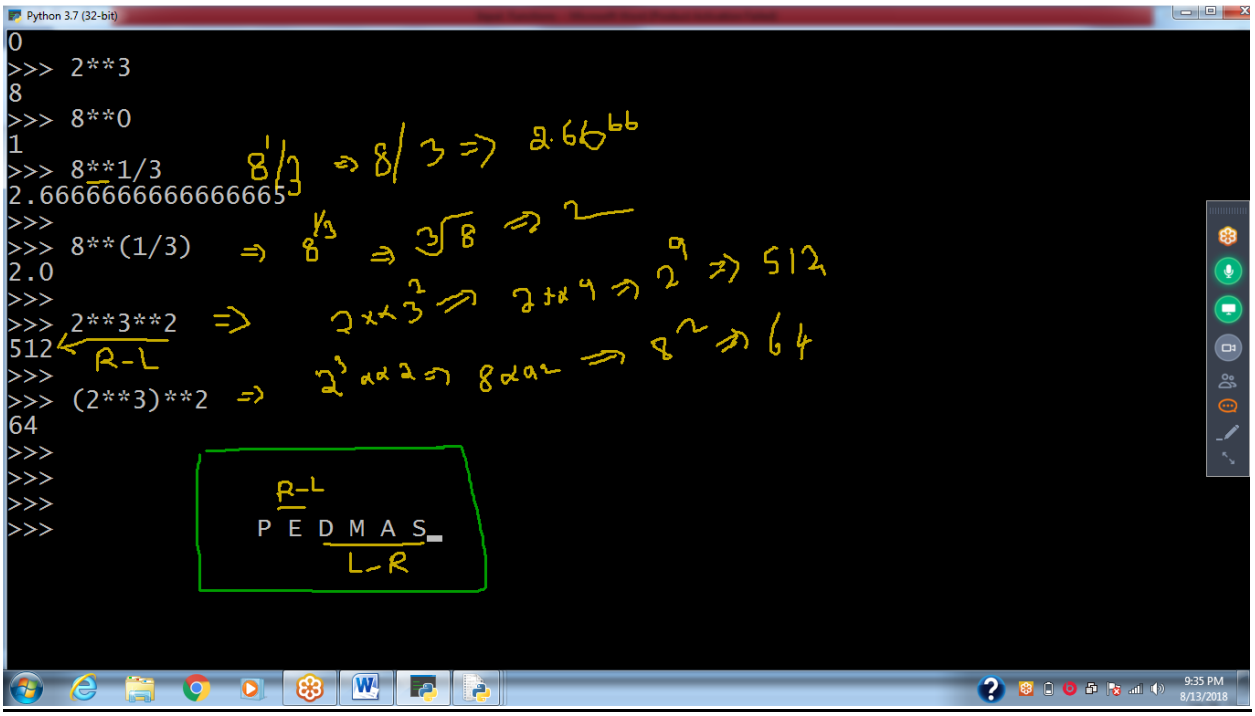
```
Python 3.7 (32-bit)
0
>>> x=input("Enter : ")
Enter : 100
>>> print(x,type(x))
100 <class 'str'>
>>>
>>> x=input("Enter : ")
Enter : 2.34
>>> print(x,type(x))
2.34 <class 'str'>
>>>
>>> x=input("Enter : ")
Enter : 2+3j
>>> print(x,type(x))
2+3j <class 'str'>
>>>
>>> x=input("Enter : ")
Enter : Hari
>>> print(x,type(x))
Hari <class 'str'>
>>> _
```

Opeartors

Arithmetic Operators :

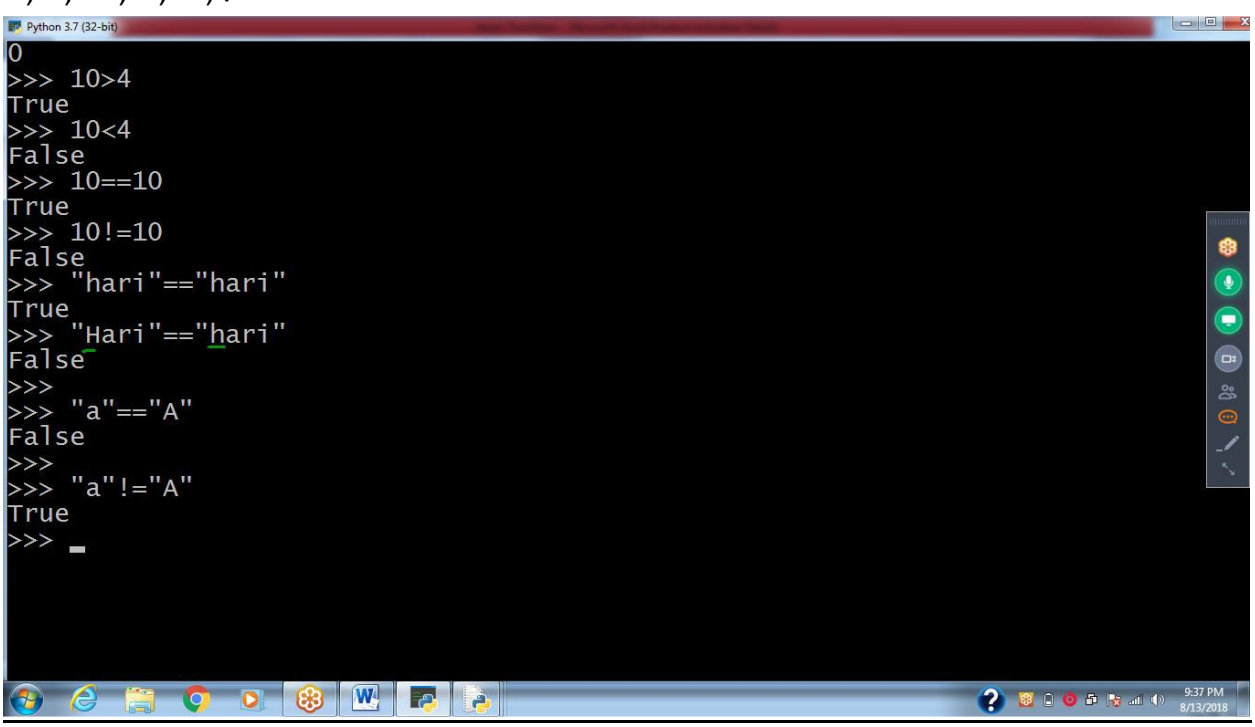
+ , - ,*, / ,%, //(Floor Division),** (exponentiation)

```
Python 3.7 (32-bit)
0
>>> 10%3
1
>>> 10%6
4
>>> 10%5
0
>>> 10/4
2.5
>>> 10/6
1.6666666666666667
>>> 10/5
2.0
>>> 10.0/4.0
2.5
>>> 10//4
2
>>> 10//6
1
>>> 10//5
2
>>> 10.0//4.0
2.0
>>>
```



Relational Operators

< , > , <= ,>=, ==, !=



Logical operators

and, or, not

not True => False

not False = > True

Assignment operator (=)

```
Python 3.7 (32-bit)
0
>>> a=100
>>> b=2.3
>>> c="Unix"
>>>
>>> a,b,c=100,2.3,"Unix"
>>>
>>> print(a,b,c)
100 2.3 Unix
>>>
>>> a,b,c=10,20,30,40
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: too many values to unpack (expected 3)
>>>
>>> a,b,c=10,20
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: not enough values to unpack (expected 3, got 2)
>>>
>>> a=10
>>> b=10
>>> c=10
>>> a=b=c=10
>>>
```

Short hand assignment operators (=)

+= , -= , *= , /= , %= , //= , **=

Eg1 : a=10

a=a+5 (or) a+=5

Eg2 :

total_marks=490

total_marks=total_marks+10

total_marks+=10

Membership Operators

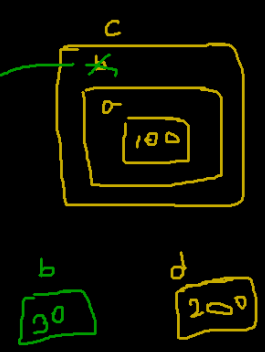
1. in
2. not in

```
Python 3.7 (32-bit)
0
>>> x="python is simple and easy to learn"
>>> "easy"   in   x
True
>>> "hard"  in   x
False
>>>
>>> "e"    in   x
True
>>>
>>> "unix"   not  in   x
True
>>>
```

Identity operators

1. is
2. is not

```
Python 3.7 (32-bit)
0
>>> a=100
>>> b=100
>>> c=a
>>> d=200
>>> print(id(a),id(b),id(c),id(d))
1565015856 1565015856 1565015856 1565017456
>>> a  is  b
True
>>> a  is  c
True
>>> a  is  d
False
>>> b=30
>>> a  is  b
False
>>> print(id(a),id(b),id(c),id(d))
1565015856 1565014736 1565015856 1565017456
>>>
```



IF_STATEMENT

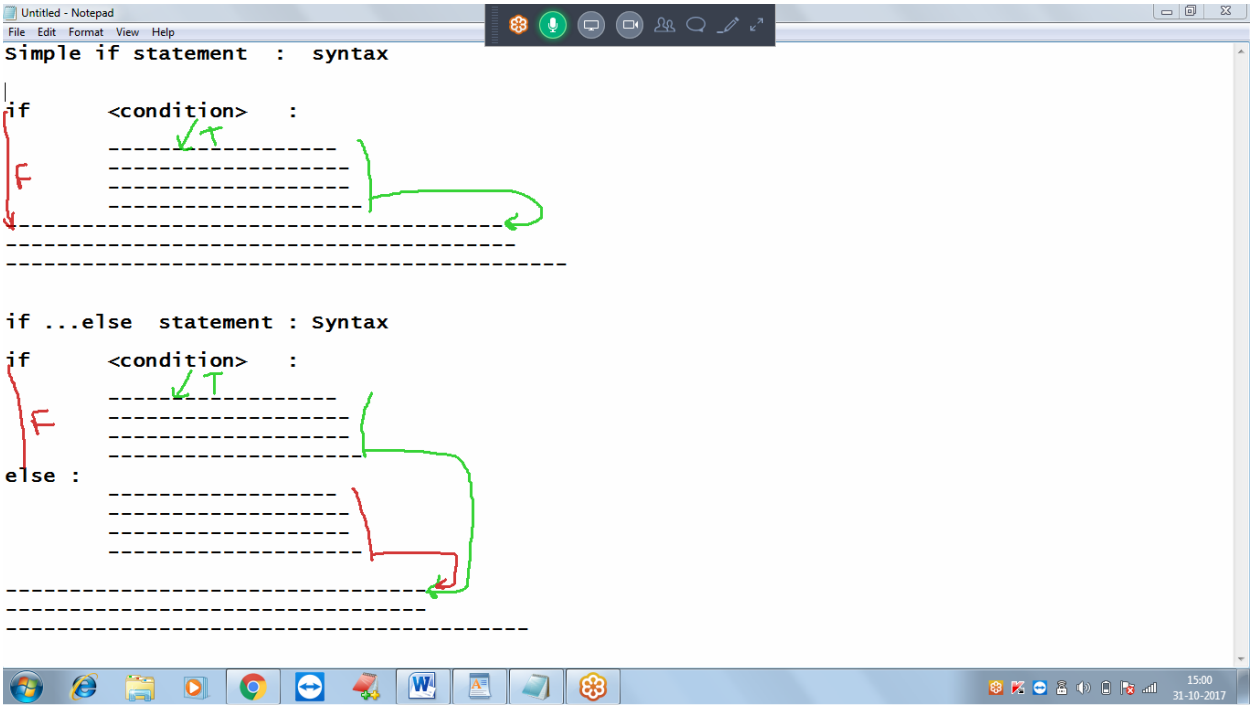
Control Statements

1. Decision Control Statements
2. Looping Control Statements

Decision Control Statements

If Statements

1. Simple if statement
2. If ….. else statement
3. If…..elif ….. statement
4. Nested if …..else statement
5. Single line if …… statement



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File Edit Format View Help

If …elif…… statement : Syntax

```
if      <condition1>  :
        ----------------------
        ----------------------
        ----------------------
elif    <condition2> :
        ----------------------
        ----------------------
        ----------------------
elif    <condition3> :
        ----------------------
        ----------------------
        ----------------------
  :
  :
elif    <condition n> :
        ----------------------
        ----------------------
        ----------------------
else :
        ----------------------
        ----------------------
        ----------------------

--------------------------------------------
--------------------------------------------
--------------------------------------------
```

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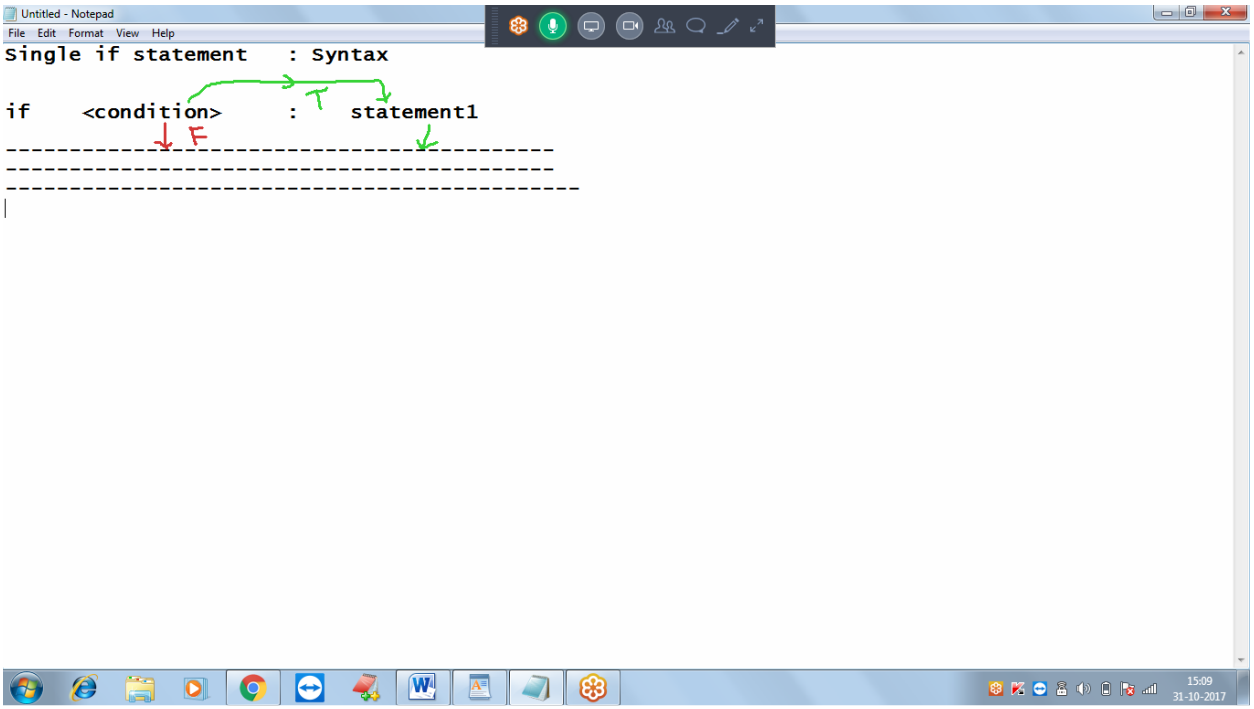
Nested if …else : Syntax

```
if    <condition 1>  :
        if    <condition 2> :
                if    <condition 3 > :
                        --------------------
                        --------------------
                        --------------------
                else  :
                        --------------------
                        --------------------
                        --------------------
        else :
                --------------------
                --------------------
                --------------------
else :
        --------------------
        --------------------
        --------------------

-------------------------------------------------------
-------------------------------------------------------
-------------------------------------------------------
```



SAMPLE PROGRAMS

==========

 # WAS accept a number check the given number is Even Or Odd number

```
n=int(input("Enter a number : "))
if  n%2==0  :
   print(n,'is an Even number')
   print('Welcome to Python class')
   print('GE to all')
else :
   print(n,'is an Odd number')
```

 # Perfect square or not

```
n=int(input("Enter a number : ")) #200
k=int(n**(1/2))
if  n==k*k :
   print(n,"is a perfect square")
```

```
else :
   print(n,"is not a perfect square")
```

 # WAS accept a number check the given number is 3 digit number

```
n=int(input("Enter a number : "))
if   abs(n)>=100  and   abs(n)<=999 :
   print(n,'is a 3 digit number')
else :
   print(n,'is not a 3 digit number')
```

 # Login program

```
uname=input("Enter a username : ")
pwd=input("Enter a password :")

if  uname=="tecno"   and  pwd=="python1" :
   print("Welcome to Python class")
else :
   print("Invalid username or password")
```

 # WAS accept a string and a character and check the given character
 # exist in the given string or not.

```
str1=input("Enter a string : ")    #  Tecnosoft
ch=input("Enter a charcater : ")  #  c
if    ch   in   str1  :
   print("Given character ",ch,"present in a given ",str1,"string.")
else :
   print("Given character ",ch,"not present in a given ",str1,"string.")
```

 # WAP program accept a number and check the given number is
 # palindrome or not

```
#  121=121, 123=321
n=input("Enter a value: ")  # 525
if   n==n[::-1] :
    print(n,"is a plaindrome ")
else :
```

```
    print(n,"is not a palindrome")
```

 # Guess number game

```
import random
x=random.randint(1,10)
n=int(input("Take a guess [1-10] : "))
if  n==x :
   print("Wow!Congratulations,you guessed number correctly.")
else:
   print("Nope,Your guess was wrong.I was thinking ",x)
```

 # Write a script accept empno,ename,job and salary and
#calculate bonus based on the following condtions

```
#job       bonus
#manager    20% on ann_sal
#analyst    18% on ann_sal
#programmer 15% on ann_sal
#salesman   12% on ann_sal
#others     10% on ann_sal

empno=int(input("Enter a  empno : "))
ename=input("Enter a ename : ")
job=input("Enter a job : ")
sal=float(input("Enter a salary : "))

ann_sal=sal*12
if job=="manager":   bonus=ann_sal*20/100
elif job=="analyst" :  bonus=ann_sal*18/100
elif job=="programmer" : bonus=ann_sal*15/100
elif job=="salesman" :  bonus=ann_sal*12/100
else :    bonus=ann_sal*10/100
print("#"   *  50)
print("###   Employee Bonus Report  ####")
print("-" * 50)
print("Employee Number  : ",empno)
print("Employee Name    : ",ename)
print("Employee Job     : ",job)
```

```
print ("mployee Salary  : ",sal)
print("Employee AnnSal  : ",ann_sal)
print("Employee Bonus   : ",bonus)
print ( "-" * 50)
print("#"   *  50)
```

 # write a script accept 2 number's and find greast number

```
a=int(input("Enter a number 1 : "))
b=int(input("Enter a number 2 : "))
if a!=b :
   if a>b :
      print(a," is big")
   else :
      print(b," is big")
else :
   print("2 numbers are equal")
```

 # If statement assignments

1) WAP accept a string and check 'a' character exist in the given string are not

2) write a script accept 2 number's and find greast number

3) WAP accept a intger number and check the given number is +ve or -ve no.

4) Write a accept a number and check the given number is multiple of 5 or not

5) WAP accept 3 integer numbers and find greatest number

6) WAP accept student name,and three subject marks and calculate result,total marks,avg marks and grade based on following conditions

NOTE : pass marks is 40

avg marks	grade
>=90	A+
>=80 and <90	A
>=70 and <80	B
>=60 and <70	C
>=50 and <60	D
<50	No grade

7)WAP accept product no,pname,price,qty and caluclate total,disc based onthe following conditions and net amount and prepare bill

Total	Discount
>10000	20% on total

>=5000 and <10000 15% on total

>=3000 and <5000 12% on total

>=1000 and < 3000 10% on total

<1000 5% on total